

Name: _____

Date: _____

Class: _____

MyDesign Portfolio

Element E Initial Template - AT - 15 points

Element E: Application of STEM Principles and Practices

STEM Principles and Practices to Design Requirements (E1)

This section should describe how your proposed solution is supported by STEM (science, technology, engineering, mathematics, computer science, etc.) concepts.

1. Concepts can be laws, theories, and practices, including disciplinary core ideas (i.e., Next Generation Science Standards, Common Core State Standards) and standards for mathematical practice.
2. Be sure to apply the applicable STEM concepts in context, directly to the specifics of your design. Review concepts learned in your other classes and tackle new concepts as appropriate. Include any relevant calculations.
3. Element C indicates that you should have at least three high-priority design requirements to receive a score of 3 (and four for a score of 4, and five for a 5). Your number of design requirements and your number of high-priority design requirements should be the same throughout your portfolio. You should use the same three to five design requirements here too, and apply **at least three of them** to STEM concepts to achieve a score of 5 on E1. Use as many copies of this sub-template as needed.

High-Priority Design Requirement #1 from Element C - Damage assessment, identifying cracks, spalling, and delamination, determining damage depth and extent, and assessing reinforcement corrosion if exposed. Damage assessment is crucial for

Name: _____

Date: _____

Class: _____

understanding how ice has affected concrete and determining the appropriate repair strategy. Ice-related damage results from de-icing chemicals, like salt, leading to various forms of deterioration such as; Cracking, Delamination, and Scaling.

You don't want your concrete falling apart after it has been laid so to keep that from happening we have to do some assessments to find the best quality to make it last long and have no issues.

STEM Concepts Involved in the requirement: Stem concepts involved science and mathematical equations in our requirements contain scientific equations with the balance of certain chemicals to see how they scale on the pH scale.

Describe how STEM concepts apply directly to the specifics of your design requirement:

These stem concepts apply directly to our design requirements as we try to find a solution from chemicals that can melt ice but have low acidic chemical balance so that it can not damage the surface of concrete.

<https://www.nextgenscience.org/three-dimensional-learning>

High-Priority Design Requirement #2 from Element C - It has to require a highly-build protective film that covers the full surface of concrete. This directly ties to our research as the film would be able to withstand and resist any sorts of abrasion or chemicals



Name: _____

Date: _____

Class: _____

meaning the salt we would pour would have no effect on the concrete while melting the ice.

STEM Concepts Involved in the requirement: Mechanical Engineering and Environmental Engineering

Describe how STEM concepts apply directly to the specifics of your design requirement:

STEM concepts apply directly to the requirement because in order for the film to do its job, we'll have to consider things like abrasion, impact resistance, and the film's effect on the environment.

High-Priority Design Requirement #3 from Element C - In order for the concrete to last many years, you have to use the right concrete sealer, that will keep the concrete from being damaged, and keep it nice looking.

The ice or salt whichever damaged the concrete shouldn't have. If they had used the right sealer the concrete would have been looking the same way it did before the snowstorm hit. For this not to happen again, we have to use the correct sealer so that the concrete can last longer and look nice.

You don't want your concrete looking bad, and being a hazard to the staff, students, and whoever visits your place. In order for this not to happen to your concrete again we have to put the right and/or best sealer down so that your concrete can look good.

STEM Concepts Involved in the requirement:

Civil Engineering and Environmental Engineering.

Describe how STEM concepts apply directly to the specifics of your design requirement:

STEM concepts apply directly to the requirement because in order to lay down the



Name: _____

Date: _____

Class: _____

sealant we have to understand which would work the best for the circumstance and we have to be able to evaluate the effect of the sealants on the environment.

High-Priority Design Requirement #4 from Element C (if you have one)-

In order to maintain a strong durable concrete concealer we would need a concrete mix of a 1:1.5:3 ratio.

It supports the idea that the more highly strengthening concrete that we would use should withstand more damage/cracks throughout the the chemicals or salts they may be used to melt the ice.

The feedback I got was that if you don't properly mix the concrete and have a wrong ratio won't the concrete be weaker and how would the ratio mix change in weather like this in the cold.

Describe how STEM concepts apply directly to the specifics of your design requirement:

- Science: The 1:1.5:3 ratio ensures optimal strength. Too much water weakens the concrete, while salts and freeze-thaw cycles cause cracks. Air-entraining agents help resist cold damage
- Technology: Proper mixing techniques and admixtures improve durability, especially in cold weather
- Engineering: A well-mixed ratio archives 25-30 MPa strength. In cold weather, using heated materials and proper curing prevents premature freezing and weakness
- Mathematics: Adjusting the water-to-cement ratio improves durability. A 1:1.5:2.5 mix in colder conditions can enhance adhesion and reduce porosity



Name: _____

Date: _____

Class: _____

High-Priority Design Requirement #5 from Element C (if you have one) -

Whatever we use to protect the concrete shouldn't damage the concrete in the long run.

Our research says that things such as chemical exposure, water damage, and freeze-thaw cycles can all damage the concrete, so we need a solution that doesn't cause any of those problems or any other problems.

We have to use the correct sealer so that your concrete won't get damaged again.

Describe how STEM concepts apply directly to the specifics of your design requirement:

- Science: Chemical exposure, water damage, and freeze-thaw cycles weaken concrete. A proper sealer prevents moisture infiltration and chemical reactions
- Technology: Sealant types must be compatible with concrete to avoid long-term damage
- Engineering: The sealer should allow breathability to prevent trapped moisture, which can cause internal cracks
- Mathematics: Sealers should have low permeability ratings and be applied at the correct thickness to ensure full coverage and durability.

Proposed Solution to STEM Principles and Practices (E2)

While E1 was focused on how STEM applies to the design requirements, E2 is focused on how STEM is used to support the design solution.

For E2, use STEM concepts to justify your design decisions. It's important that the STEM Principles and Practices that are discussed are actually appropriate and useful to the problem. Use as many copies of this sub-template as needed.

STEM Principles for Design Solution #1 - Heated Walkway

STEM Concept: Heated Technologies, Energy Engineering, Electrical Engineering, Geometry and Layout.



Name: _____

Date: _____

Class: _____

Describe how STEM concept applies directly to the specifics of the first solution to your problem: STEM concepts apply directly to the specifics of the solution because we have to be able to figure out how we want to use the heat and how much energy we'll need from the heat to melt the ice.

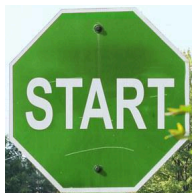


You can only work on the next sections after you have built and tested your first prototype. You need to skip to the "Evidence is Reviewed" section.

STEM Principles for Design Solution #1 - Heated Walkway

STEM Concept: Heated Technologies, Energy Engineering, Electrical Engineering, Geometry and Layout.

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Name: _____

Date: _____

Class: _____

Evidence Design is Reviewed (E3)

"Field experts" are professionals who spend at least a portion of their time working in a field directly related to your project. These field experts are not permitted to double count as stakeholders.

E3 is about experts reviewing what you've said about the STEM connections in E1 and E2. It's important here to talk to an appropriate field expert who knows about the STEM content of the project - not to a stakeholder.

Documentation #1 (Only one documentation section needed for full credit):

What format: Notes from presentation

- The expert said that the connection from our design requirements made sense.
- The expert said the practices substantiated the solution.
- The expert said we should use less aggregate and is trying to resurface the concrete since there is water inside the aggregate that can make the concrete surface easier to be dissolved.
-

Experts with Credentials (E4)

Document who your field experts are (names are not necessary) and what qualifies them as an expert. Use as many copies of this sub-template as needed.

Expert #1 (from documentation 1): Mr. Butler was the expert and what qualifies him as an expert is his role in engineering.

Verification or Corrective Response to Design Review (E5)

Your conversation/interview with the expert(s) should inform some corrective responses. In E5, describe how those reviews informed your responses.

Name: _____

Date: _____

Class: _____

For example,

"Mr Butler _____ told us _____, which informed our decision to change/reconsider _____." (or "...which verified/reinforced our decision to _____.")

Make sure to document the experts' reviews (suggestions/feedback or criticisms of your proposed solution) and how they will inform iterations you will make to your proposed solution. Use as many copies of this sub-template as needed.

Interview with Expert #1 informs corrective response:

Interview with Expert #2 informs corrective response:

Interview with Expert #3 informs corrective response:

	Criteria	5	4	3	2	1	0
	3. Evidence that Design is Reviewed. There is _____ evidence that the application of STEM principles & practices (or suitable alternative) has been reviewed by expert(s).	<i>substantial</i>	<i>adequate</i>	<i>some</i>	<i>minimal</i>	<i>no - but review was requested -</i>	<i>no</i>



Name: _____

Date: _____

Class: _____

4.Design Reviewed by Experts. _____ expert(s) have/has reviewed the design for application of STEM principles & practices.	<i>Two or more, with at least two with documented, relevant credentials</i>	<i>Two or more, with at least one with documented, relevant credentials</i>	<i>At least one with documented, clearly relevant credentials</i>	<i>At least one with documented credentials</i>	<i>Only one with unclear credentials</i>	<i>No</i>
5.Design Review Results in Verification. The design reviews provided _____.	<i>clear and detailed confirmation (verification) or substantial detail necessary to inform a corrective response</i>	<i>clear confirmation (verification) or detail necessary to inform a corrective response</i>	<i>partial confirmation (verification) or some detail necessary to inform a corrective response</i>	<i>does not provide clear confirmation (verification) or suggestions for a corrective response, but at least some response is provided</i>	<i>show no evidence of confirmation (verification) or suggestions for a corrective response by an expert, but includes notes of attempts to gain feedback</i>	<i>show no evidence of confirmation (verification) or suggestions for a corrective response by an expert</i>

